

How many Boolean functions?

How many Boolean functions (or connectives) with n arguments we have?

- $n = 0 \rightarrow 2$

f_T
1

f_F
0

- $n = 1 \rightarrow 4$

A	f_{id}
0	0
1	1

A	$f_{\neg}$
0	1
1	0

A	$f_{1,1}$
0	1
1	1

A	$f_{0,1}$
0	0
1	0

How many Boolean functions?

How many Boolean functions (or connectives) with n arguments we have?

In general, all the truth tables of the various Boolean functions with the same number n of arguments have a number of rows equal to the number of interpretations that we can define on n propositional variables, i.e., 2^n . Two such truth tables differ only in the last columns. Now, since there are 2^{2^n} possible columns of values 0,1 with 2^n rows, it follows that there are 2^{2^n} possible, Boolean functions with n arguments.

- $n = 2 \rightarrow 16$

the number of tables with all possible combinations of $\{0, 1\}$ in 2 columns, i.e., number of tables with $2^2 = 4$ rows, and 2 + 1 columns, i.e., the number of $\{0, 1\}$ vectors of 4 elements, i.e., $2^4 = 16$.

- n generic $\rightarrow 2^{2^n}$

the number of tables with all possible combinations of $\{0, 1\}$ in n columns, i.e., number of tables with 2^n rows, and $n + 1$ columns, i.e., the number of $\{0, 1\}$ vectors of 2^n elements, i.e., 2^{2^n} .

Truth tables for formulas

It is immediate to build a **truth table for a formula**: if the formula has n propositional variables, the table has one row for each interpretation of the formula (assigning truth values to all the propositional variables), and $n + 1$ columns, where for a row corresponding to an interpretation $\mathcal{I}$, the last column will contain $\mathcal{I}(A)$.

Example

Truth table for $(F \vee G) \wedge \neg(F \wedge G)$.

F	G	$(F \vee G) \wedge \neg(F \wedge G)$
True	True	False
True	False	True
False	True	True
False	False	False

Note: intuitively, what does the formula in the example represent?

Truth tables for formulas: exercises

Suggestion: to simplify the process of building a truth table for a formula, we can add columns to the table corresponding to the various subformulas

Example: Truth table for $(F \vee G) \wedge \neg(F \wedge G)$

F	G	$F \vee G$	$F \wedge G$	$\neg(F \wedge G)$	$(F \vee G) \wedge \neg(F \wedge G)$
True	True	True	True	False	False
True	False	True	False	True	True
False	True	True	False	True	True
False	False	False	False	True	False

Exercise: build the truth tables for the following propositional formulas:

- $(p \rightarrow p) \rightarrow p$
- $p \rightarrow (p \rightarrow p)$
- $p \vee q \rightarrow p \wedge q$
- $p \vee (q \wedge r) \rightarrow (p \wedge r) \vee q$
- $p \rightarrow (q \rightarrow p)$
- $(p \wedge \neg q) \vee \neg(p \leftrightarrow q)$

From formulas to Boolean functions

It can be shown that there is total correspondence between Boolean functions and formulas. One direction is easy to prove.

Proposition

For every formula A with n propositional variables $p_1, \dots, p_n$, one can define a Boolean function f_A with n arguments such that for every interpretation $\mathcal{I}$, $\mathcal{I}(A) = f_A(\mathcal{I}(p_1), \dots, \mathcal{I}(p_n))$.

The proof of the above proposition is based on the fact that the truth table for a formula directly specifies the corresponding Boolean function.

From Boolean functions to formulas

It can be shown that there is total correspondence between Boolean functions and formulas. The other direction is represented by the following proposition.

Proposition

For every Boolean function f with n arguments, there is a corresponding formula A_f with n propositional variables $p_1, \dots, p_n$, such that for every interpretation $\mathcal{I}$, $\mathcal{I}(A_f) = f(\mathcal{I}(p_1), \dots, \mathcal{I}(p_n))$.

The above proposition is more difficult to prove.

From Boolean functions to formulas: first intuition

Given the truth table of a Boolean function ϕ with n arguments ($n > 0$), we can build the formula with $n + 1$ propositional variables $p_0, p_1, \dots, p_n$ of the form

$$(p_0 \wedge \neg p_0) \vee \alpha_1 \vee \dots \vee \alpha_k$$

with one α_i for each row with value 1 for ϕ , where each α_i has the form

$$\gamma_1 \wedge \dots \wedge \gamma_n$$

where γ_j is p_j if the row has 1 in column j , and is $\neg p_j$ if the row has value 0 in column j .

Note that $(p_0 \wedge \neg p_0)$ takes care of the case where $\phi(p_1, \dots, p_n) = 0$.

Example: formula corresponding to the Boolean function ϕ

p_1	p_2	p_3	$\phi(p_1, p_2, p_3)$
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0



$$(p_0 \wedge \neg p_0) \vee (\neg p_1 \wedge \neg p_2 \wedge \neg p_3) \vee (\neg p_1 \wedge p_2 \wedge p_3)$$

From Boolean functions to formulas: second intuition

Given the truth table of a Boolean function ψ with n arguments ($n > 0$), we can build the formula with $n + 1$ propositional variables $p_0, p_1, \dots, p_n$ of the form

$$\neg((p_0 \wedge \neg p_0) \vee \beta_1 \vee \dots \vee \beta_k)$$

with one β_i for each row with value 0 for ψ , where each β_i has the form

$$\gamma_1 \wedge \dots \wedge \gamma_n$$

where γ_j is p_j if the row has 1 in column j , and is $\neg p_j$ if the row has value 0 in column j .

Note that $(p_0 \wedge \neg p_0)$ takes care of the case where $\phi(p_1, \dots, p_n) = 1$.

Example: formula corresponding to the Boolean function ψ

p_1	p_2	p_3	$\psi(p_1, p_2, p_3)$
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1



$$\neg((p_0 \wedge \neg p_0) \vee (\neg p_1 \wedge \neg p_2 \wedge p_3) \vee (p_1 \wedge \neg p_2 \wedge \neg p_3))$$

by De Morgan, equivalent to

$$(p_0 \vee \neg p_0) \wedge (p_1 \vee p_2 \vee \neg p_3) \wedge (\neg p_1 \vee p_2 \vee p_3)$$

Complete sets of connectives

The two intuitions above are at the basis of the proof that every Boolean function can be represented by a formula. Such proof shows that the following proposition is true.

Proposition

Every formula of propositional logic can be written using only the connectives $\neg$, $\vee$ and $\wedge$.

In other words, the set of connectives $\{\neg, \vee, \wedge\}$ is **complete**, because they allow to express any formula that corresponds to a Boolean function.

Note that this implies that the set of our connectives, i.e., $\{\neg, \vee, \wedge, \rightarrow, \equiv\}$ is also complete.

Even more specifically, it can be shown that at least two *normal forms* exist for propositional formulas.

Normal form for formulas

Proposition

Every Boolean function can be expressed by a formula of propositional logic in the so-called *disjunctive normal form*, i.e., as

$$\alpha_1 \vee \cdots \vee \alpha_k$$

where each α_i is a *minterm*, i.e., has the form

$$\gamma_1 \wedge \cdots \wedge \gamma_n$$

where each γ_j is a *literal*, i.e., either a propositional variable or the negation of a propositional variable.

Proposition

Every Boolean function can be expressed by a formula of propositional logic in the so-called *conjunctive normal form*, i.e., as

$$\beta_1 \wedge \cdots \wedge \beta_k$$

where each β_i is a *clause*, i.e., has the form

$$\gamma_1 \vee \cdots \vee \gamma_n$$

where each γ_j is a *literal*.

Are all our connectives necessary?

We leave as an exercise to prove the following propositions, regarding other complete sets of connectives.

Proposition

Every formula of propositional logic can be written using only the connectives $\neg$ and $\vee$.

Proposition

Every formula of propositional logic can be written using only the connectives $\neg$ and $\wedge$.

Also, if we assume we can use also T and F in formulas, the following propositions hold.

Proposition

Every formula of propositional logic can be written using only the connective NAND or only the connective NOR.

Valid, satisfiable, and unsatisfiable formulas

Definition (Model of a formula)

An interpretation $\mathcal{I}$ that satisfies a formula A , i.e., such that $\mathcal{I} \models A$, is called a **model** of A .

Definition

A formula A is

Valid or **tautology** if **for all interpretations** $\mathcal{I}$, $\mathcal{I} \models A$
i.e., every interpretation is a model of A

Satisfiable if **there is an interpretation** $\mathcal{I}$ s.t. $\mathcal{I} \models A$
i.e., A has at least one model

Unsatisfiable if **for no interpretation** $\mathcal{I}$, $\mathcal{I} \models A$
i.e., A has no model

Valid, satisfiable, and unsatisfiable formulas

Proposition

$A \text{ valid} \longrightarrow A \text{ satisfiable} \longleftrightarrow A \text{ not unsatisfiable}$

$A \text{ unsatisfiable} \longleftrightarrow A \text{ not satisfiable} \longrightarrow A \text{ not Valid}$

Proposition

<i>if A is</i>	<i>then $\neg A$ is</i>
<i>Valid</i>	<i>Unsatisfiable</i>
<i>Satisfiable</i>	<i>not Valid</i>
<i>not Valid</i>	<i>Satisfiable</i>
<i>Unsatisfiable</i>	<i>Valid</i>

Checking validity and (un)satisfiability of a formula

Truth table

Checking (un)satisfiability and validity of a formula A can be done by enumerating all the interpretations which are relevant for A , and for each interpretation $\mathcal{I}$ check if $\mathcal{I} \models A$.

Example (using truth table)

A	B	C	$A \rightarrow (B \vee \neg C)$
True	True	True	True
True	True	False	True
True	False	True	False
True	False	False	True
False	True	True	True
False	True	False	True
False	False	True	True
False	False	False	True

Valid, satisfiable, and unsatisfiable formulas

Example

Satisfiable	$A \rightarrow A$	}	Valid
	$A \vee \neg A$		
	$\neg\neg A \equiv A$		
	$\neg(A \wedge \neg A)$		
	$A \wedge B \rightarrow A$		
	$A \rightarrow A \vee B$		
Unsatisfiable	$A \vee B$	}	Non Valid
	$A \rightarrow B$		
	$\neg(A \vee B) \rightarrow C$		
	$A \wedge \neg A$		
	$\neg(A \rightarrow A)$		
	$A \equiv \neg A$		
	$\neg(A \equiv A)$		

Prove that the blue formulas are valid, that the magenta formulas are satisfiable but not valid, and that the red formulas are unsatisfiable.